

CoP SAP 6.2 Appendix

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July 24, 2026

6.2 Appendix Correlations of Virus-Specific Antibodies and Viral Load by Sex

We recalculate some of the Tables from CoP SAP 6.2 by sex.

Virus-Specific Antibodies

Correlations

Simple statistics will be performed independent of the vaccine product. For example, the Spearman correlation between each pair of virus-specific antibodies will be calculated. Spearman (i.e., rank) correlations with confidence intervals will be generated; confidence intervals for the Spearman's correlation coefficient are calculated by using the ranks in a Pearson's correlation, which generally performs well (Caruso and Cliff, 1997). As there were not many ties, nonparametric bootstrap as a sensitivity analysis was not done.

Table 6.2.1m: (Male Only) Spearman Correlations of Virus-Specific Antibodies

	log MN 2 weeks prior chal- lenge	log MN day chal- lenge	log MSD ECL 2 weeks prior chal- lenge	log MSD ECL day chal- lenge	log PsVNA ID50 2 weeks prior chal- lenge	log PsVNA ID50 day chal- lenge	log PsVNA ID80 2 weeks prior chal- lenge	log PsVNA ID80 day chal- lenge
log MN 2 weeks prior challenge	-	-	-	-	-	-	-	-
log MN day challenge	0.89 (0.82, 0.93)	-	-	-	-	-	-	-
log MSD ECL 2 weeks prior challenge	0.92 (0.87, 0.95)	0.89 (0.83, 0.94)	-	-	-	-	-	-
log MSD ECL day challenge	0.91 (0.86, 0.95)	0.9 (0.84, 0.94)	0.98 (0.97, 0.99)	-	-	-	-	-
log PsVNA ID50 2 weeks prior challenge	0.91 (0.86, 0.95)	0.87 (0.79, 0.92)	0.92 (0.87, 0.95)	0.91 (0.85, 0.94)	-	-	-	-
log PsVNA ID50 day challenge	0.89 (0.82, 0.93)	0.84 (0.76, 0.9)	0.91 (0.85, 0.94)	0.9 (0.83, 0.94)	0.94 (0.9, 0.96)	-	-	-
log PsVNA ID80 2 weeks prior challenge	0.93 (0.89, 0.96)	0.87 (0.79, 0.92)	0.93 (0.89, 0.96)	0.91 (0.86, 0.95)	0.97 (0.95, 0.98)	0.92 (0.87, 0.95)	-	-
log PsVNA ID80 day challenge	0.93 (0.88, 0.96)	0.9 (0.84, 0.94)	0.94 (0.9, 0.96)	0.94 (0.91, 0.97)	0.95 (0.91, 0.97)	0.94 (0.9, 0.96)	0.95 (0.92, 0.97)	-

6.2.1f: (Female Only) Spearman Correlations of Virus-Specific Antibodies

	log MN 2 weeks prior chal- lenge	log MN day chal- lenge	log MSD ECL 2 weeks prior chal- lenge	log MSD ECL day chal- lenge	log PsVNA ID50 2 weeks prior chal- lenge	log PsVNA ID50 day chal- lenge	log PsVNA ID80 2 weeks prior chal- lenge	log PsVNA ID80 day chal- lenge
log MN 2 weeks prior challenge	-	-	-	-	-	-	-	-
log MN day challenge	0.91 (0.85, 0.94)	-	-	-	-	-	-	-
log MSD ECL 2 weeks prior challenge	0.93 (0.89, 0.96)	0.88 (0.81, 0.93)	-	-	-	-	-	-
log MSD ECL day challenge	0.92 (0.88, 0.95)	0.88 (0.81, 0.93)	0.97 (0.95, 0.98)	-	-	-	-	-
log PsVNA ID50 2 weeks prior challenge	0.88 (0.81, 0.93)	0.85 (0.77, 0.91)	0.9 (0.84, 0.94)	0.87 (0.79, 0.92)	-	-	-	-
log PsVNA ID50 day challenge	0.87 (0.79, 0.92)	0.87 (0.8, 0.92)	0.89 (0.82, 0.93)	0.88 (0.81, 0.93)	0.92 (0.87, 0.95)	-	-	-
log PsVNA ID80 2 weeks prior challenge	0.91 (0.86, 0.95)	0.88 (0.82, 0.93)	0.93 (0.89, 0.96)	0.92 (0.86, 0.95)	0.95 (0.92, 0.97)	0.92 (0.87, 0.95)	-	-
log PsVNA ID80 day challenge	0.92 (0.88, 0.95)	0.93 (0.88, 0.95)	0.92 (0.87, 0.95)	0.93 (0.89, 0.96)	0.92 (0.86, 0.95)	0.94 (0.91, 0.96)	0.94 (0.9, 0.96)	-

Viral Load

Correlations - Numeric Viral Load

Spearman's correlations of all numeric viral load measurements with each other will be calculated.

6.2.2m: (Male Only) Spearman Correlations of Viral Load Outcomes

	AUC BAL N1	AUC BAL Subge- nomic	AUC Nasal Swab N1	AUC Nasal Swab Subge- nomic	AUC OP Swab N1	AUC OP Swab Subge- nomic	log VL Day2 Post Chal- lenge BAL N1	log VL Day2 Post Chal- lenge BAL Subge- nomic	log VL Day2 Post Chal- lenge OP- swab N1	log VL Day2 Post Chal- lenge OP- swab Subge- nomic	log VL Day2 Post Chal- lenge Nasal- Swab N1	log VL Day2 Post Chal- lenge Nasal- Swab Subge- nomic
AUC BAL N1	-	-	-	-	-	-	-	-	-	-	-	-
AUC BAL Subgenomic	0.91 (0.86, 0.95)	-	-	-	-	-	-	-	-	-	-	-
AUC Nasal Swab N1	0.63 (0.45, 0.76)	0.61 (0.43, 0.75)	-	-	-	-	-	-	-	-	-	-
AUC Nasal Swab Subgenomic	0.6 (0.42, 0.74)	0.57 (0.37, 0.72)	0.83 (0.74, 0.89)	-	-	-	-	-	-	-	-	-
AUC OP Swab N1	0.8 (0.69, 0.88)	0.75 (0.62, 0.84)	0.8 (0.69, 0.87)	0.71 (0.56, 0.81)	-	-	-	-	-	-	-	-
AUC OP Swab Subgenomic	0.74 (0.6, 0.83)	0.69 (0.53, 0.8)	0.69 (0.54, 0.8)	0.63 (0.46, 0.76)	0.89 (0.83, 0.93)	-	-	-	-	-	-	-
log VL Day2 PostChal- lenge BAL N1	0.88 (0.81, 0.93)	0.91 (0.85, 0.94)	0.54 (0.34, 0.69)	0.49 (0.28, 0.66)	0.7 (0.55, 0.81)	0.6 (0.42, 0.74)	-	-	-	-	-	-
log VL Day2 PostChal- lenge BAL Subgenomic	0.84 (0.74, 0.9)	0.94 (0.9, 0.96)	0.54 (0.33, 0.69)	0.49 (0.28, 0.66)	0.68 (0.52, 0.8)	0.6 (0.41, 0.74)	0.95 (0.91, 0.97)	-	-	-	-	-
log VL Day2 PostChal- lenge OPswab N1	0.46 (0.24, 0.63)	0.4 (0.17, 0.59)	0.38 (0.15, 0.58)	0.43 (0.21, 0.61)	0.52 (0.32, 0.68)	0.55 (0.35, 0.7)	0.35 (0.11, 0.55)	0.34 (0.1, 0.54)	-	-	-	-
log VL Day2 PostChal- lenge OPswab Subgenomic	0.46 (0.24, 0.63)	0.42 (0.19, 0.61)	0.42 (0.19, 0.6)	0.48 (0.26, 0.65)	0.57 (0.37, 0.72)	0.61 (0.43, 0.75)	0.34 (0.1, 0.54)	0.35 (0.12, 0.55)	0.96 (0.93, 0.97)	-	-	-

log VL Day2	0.57	0.56	0.69	0.77	0.73	0.6	0.54	0.55	0.48	0.52	-	-
PostChal-	(0.37,	(0.37,	(0.53,	(0.64,	(0.59,	(0.41,	(0.34,	(0.35,	(0.26,	(0.31,		
lenge	0.71)	0.71)	0.8)	0.85)	0.83)	0.74)	0.7)	0.7)	0.65)	0.68)		
NasalSwab												
N1												
log VL Day2	0.54	0.56	0.68	0.79	0.7	0.59	0.53	0.56	0.44	0.49	0.97	-
PostChal-	(0.34,	(0.36,	(0.51,	(0.67,	(0.55,	(0.41,	(0.32,	(0.36,	(0.22,	(0.28,	(0.95,	
lenge	0.7)	0.71)	0.79)	0.87)	0.81)	0.73)	0.69)	0.71)	0.62)	0.66)	0.98)	
NasalSwab												
Subgenomic												

6.2.2f: (Female Only) Spearman Correlations of Viral Load Outcomes

	AUC BAL N1	AUC BAL Subge- nomic	AUC Nasal Swab N1	AUC Nasal Swab Subge- nomic	AUC OP Swab N1	AUC OP Swab Subge- nomic	log VL Day2 Post Chal- lenge BAL N1	log VL Day2 Post Chal- lenge BAL Subge- nomic	log VL Day2 Post Chal- lenge OP- swab N1	log VL Day2 Post Chal- lenge OP- swab Subge- nomic	log VL Day2 Post Chal- lenge Nasal- Swab N1	log VL Day2 Post Chal- lenge Nasal- Swab Subge- nomic
AUC BAL N1	-	-	-	-	-	-	-	-	-	-	-	-
AUC BAL Subgenomic	0.83 (0.73, 0.89)	-	-	-	-	-	-	-	-	-	-	-
AUC Nasal Swab N1	0.61 (0.42, 0.74)	0.59 (0.4, 0.73)	-	-	-	-	-	-	-	-	-	-
AUC Nasal Swab Subgenomic	0.61 (0.43, 0.75)	0.52 (0.32, 0.68)	0.89 (0.83, 0.93)	-	-	-	-	-	-	-	-	-
AUC OP Swab N1	0.66 (0.49, 0.78)	0.49 (0.27, 0.66)	0.76 (0.63, 0.85)	0.79 (0.67, 0.87)	-	-	-	-	-	-	-	-
AUC OP Swab Subgenomic	0.66 (0.49, 0.78)	0.52 (0.31, 0.68)	0.71 (0.56, 0.82)	0.74 (0.6, 0.84)	0.9 (0.84, 0.94)	-	-	-	-	-	-	-
log VL Day2 PostChal- lenge BAL N1	0.85 (0.76, 0.91)	0.83 (0.74, 0.9)	0.44 (0.22, 0.62)	0.45 (0.23, 0.63)	0.47 (0.25, 0.65)	0.51 (0.3, 0.68)	-	-	-	-	-	-
log VL Day2 PostChal- lenge BAL Subgenomic	0.8 (0.69, 0.88)	0.92 (0.88, 0.95)	0.45 (0.23, 0.63)	0.44 (0.21, 0.62)	0.45 (0.22, 0.63)	0.48 (0.26, 0.65)	0.9 (0.84, 0.94)	-	-	-	-	-
log VL Day2 PostChal- lenge OPswab N1	0.41 (0.18, 0.6)	0.31 (0.07, 0.52)	0.52 (0.32, 0.68)	0.47 (0.25, 0.64)	0.53 (0.33, 0.69)	0.61 (0.43, 0.75)	0.28 (0.03, 0.5)	0.19 (- 0.06, 0.42)	-	-	-	-
log VL Day2 PostChal- lenge OPswab Subgenomic	0.46 (0.24, 0.63)	0.37 (0.14, 0.57)	0.59 (0.41, 0.73)	0.52 (0.31, 0.68)	0.55 (0.35, 0.7)	0.66 (0.5, 0.78)	0.33 (0.08, 0.53)	0.25 (0, 0.47)	0.97 (0.95, 0.98)	-	-	-
log VL Day2 PostChal- lenge NasalSwab N1	0.53 (0.33, 0.69)	0.39 (0.16, 0.58)	0.74 (0.6, 0.83)	0.75 (0.62, 0.84)	0.62 (0.44, 0.75)	0.61 (0.42, 0.74)	0.38 (0.15, 0.58)	0.29 (0.05, 0.51)	0.49 (0.28, 0.66)	0.56 (0.37, 0.71)	-	-

log VL Day2	0.55	0.4	0.74	0.79	0.64	0.64	0.37	0.29	0.51	0.58	0.98	-
PostChal-	(0.35,	(0.17,	(0.6,	(0.68,	(0.47,	(0.47,	(0.14,	(0.05,	(0.3,	(0.39,	(0.96,	
lenge	0.7)	0.59)	0.83)	0.87)	0.77)	0.77)	0.57)	0.51)	0.67)	0.73)	0.99)	
NasalSwab												
Subgenomic												

Virus-Specific Antibodies by Viral Load

For numeric viral load, Spearman correlations with each virus-specific Ab will be calculated.

6.2.3m: (Male Only) Spearman Correlations of Virus-Specific Antibodies with Viral Load Measurements

	log MN 2 weeks prior chal- lenge	log MN day chal- lenge	log MSD ECL 2 weeks prior chal- lenge	log MSD ECL day chal- lenge	log PsVNA ID50 2 weeks prior chal- lenge	log PsVNA ID50 day chal- lenge	log PsVNA ID80 2 weeks prior chal- lenge	log PsVNA ID80 day chal- lenge
AUC BAL	-0.53	-0.67	-0.58	-0.61	-0.54	-0.5	-0.58	-0.6
N1	(-0.69, -0.32)	(-0.79, -0.51)	(-0.73, -0.39)	(-0.74, -0.42)	(-0.7, -0.34)	(-0.66, -0.28)	(-0.72, -0.39)	(-0.74, -0.41)
AUC BAL	-0.51	-0.62	-0.55	-0.59	-0.51	-0.44	-0.56	-0.57
Subgenomic	(-0.68, -0.31)	(-0.76, -0.45)	(-0.7, -0.35)	(-0.73, -0.4)	(-0.67, -0.3)	(-0.62, -0.22)	(-0.71, -0.36)	(-0.72, -0.38)
AUC Nasal	-0.69	-0.66	-0.73	-0.74	-0.67	-0.62	-0.68	-0.74
Swab N1	(-0.8, -0.54)	(-0.78, -0.49)	(-0.83, -0.59)	(-0.83, -0.6)	(-0.79, -0.51)	(-0.75, -0.44)	(-0.8, -0.52)	(-0.83, -0.6)
AUC Nasal	-0.72	-0.73	-0.73	-0.74	-0.73	-0.64	-0.72	-0.76
Swab	(-0.82, -0.57)	(-0.82, -0.58)	(-0.83, -0.59)	(-0.83, -0.6)	(-0.83, -0.59)	(-0.77, -0.47)	(-0.82, -0.57)	(-0.85, -0.63)
Subgenomic								
AUC OP	-0.65	-0.7	-0.72	-0.73	-0.64	-0.58	-0.68	-0.7
Swab N1	(-0.77, -0.47)	(-0.8, -0.54)	(-0.82, -0.57)	(-0.83, -0.59)	(-0.77, -0.47)	(-0.73, -0.39)	(-0.79, -0.51)	(-0.81, -0.54)
AUC OP	-0.56	-0.61	-0.6	-0.64	-0.57	-0.49	-0.6	-0.6
Swab	(-0.71, -0.36)	(-0.74, -0.43)	(-0.74, -0.41)	(-0.77, -0.47)	(-0.72, -0.38)	(-0.66, -0.28)	(-0.74, -0.42)	(-0.74, -0.41)
Subgenomic								
log VL Day2	-0.41	-0.53	-0.46	-0.49	-0.43	-0.38	-0.46	-0.49
PostChal- lenge BAL	(-0.59, -0.18)	(-0.69, -0.33)	(-0.63, -0.24)	(-0.66, -0.28)	(-0.61, -0.21)	(-0.58, -0.15)	(-0.63, -0.24)	(-0.66, -0.27)
N1								
log VL Day2	-0.45	-0.54	-0.48	-0.51	-0.46	-0.4	-0.5	-0.51
PostChal- lenge BAL	(-0.63, -0.23)	(-0.7, -0.34)	(-0.65, -0.26)	(-0.67, -0.3)	(-0.63, -0.24)	(-0.59, -0.17)	(-0.66, -0.29)	(-0.67, -0.3)
Subgenomic								
log VL Day2	-0.39	-0.4	-0.36	-0.39	-0.41	-0.36	-0.42	-0.41
PostChal- lenge	(-0.58, -0.16)	(-0.59, -0.16)	(-0.56, -0.13)	(-0.58, -0.15)	(-0.6, -0.18)	(-0.56, -0.13)	(-0.6, -0.19)	(-0.6, -0.18)
OPswab N1								
log VL Day2	-0.45	-0.45	-0.43	-0.45	-0.48	-0.43	-0.49	-0.47
PostChal- lenge	(-0.63, -0.23)	(-0.63, -0.23)	(-0.61, -0.2)	(-0.63, -0.23)	(-0.65, -0.26)	(-0.61, -0.21)	(-0.66, -0.27)	(-0.64, -0.25)
OPswab								
Subgenomic								
log VL Day2	-0.78	-0.77	-0.8	-0.77	-0.82	-0.75	-0.81	-0.82
PostChal- lenge	(-0.86, -0.67)	(-0.85, -0.64)	(-0.87, -0.68)	(-0.86, -0.65)	(-0.89, -0.71)	(-0.84, -0.62)	(-0.88, -0.7)	(-0.89, -0.72)
NasalSwab								
N1								

log VL Day2	-0.77	-0.76	-0.77	-0.76	-0.8	-0.74	-0.79	-0.81
PostChal-	(-0.85,	(-0.85,	(-0.85,	(-0.84,	(-0.87,	(-0.83,	(-0.87,	(-0.88,
lenge	-0.64)	-0.63)	-0.64)	-0.62)	-0.68)	-0.6)	-0.67)	-0.7)
NasalSwab								
Subgenomic								

6.2.3f: (Female Only) Spearman Correlations of Virus-Specific Antibodies with Viral Load Measurements

	log MN 2 weeks prior chal- lenge	log MN day chal- lenge	log MSD ECL 2 weeks prior chal- lenge	log MSD ECL day chal- lenge	log PsVNA ID50 2 weeks prior chal- lenge	log PsVNA ID50 day chal- lenge	log PsVNA ID80 2 weeks prior chal- lenge	log PsVNA ID80 day chal- lenge
AUC BAL	-0.72	-0.75	-0.71	-0.72	-0.71	-0.68	-0.68	-0.71
N1	(-0.82, -0.58)	(-0.84, -0.62)	(-0.82, -0.57)	(-0.82, -0.58)	(-0.81, -0.56)	(-0.8, -0.53)	(-0.8, -0.52)	(-0.81, -0.56)
AUC BAL	-0.63	-0.63	-0.63	-0.63	-0.58	-0.55	-0.56	-0.56
Subgenomic	(-0.76, -0.45)	(-0.76, -0.46)	(-0.76, -0.46)	(-0.76, -0.45)	(-0.73, -0.39)	(-0.7, -0.35)	(-0.71, -0.37)	(-0.71, -0.37)
AUC Nasal	-0.7	-0.71	-0.68	-0.72	-0.6	-0.67	-0.65	-0.72
Swab N1	(-0.8, -0.54)	(-0.82, -0.57)	(-0.8, -0.52)	(-0.82, -0.57)	(-0.74, -0.41)	(-0.79, -0.51)	(-0.77, -0.48)	(-0.82, -0.58)
AUC Nasal	-0.71	-0.72	-0.68	-0.7	-0.62	-0.65	-0.67	-0.72
Swab	(-0.82, -0.57)	(-0.82, -0.58)	(-0.8, -0.53)	(-0.81, -0.55)	(-0.75, -0.44)	(-0.77, -0.48)	(-0.79, -0.51)	(-0.82, -0.58)
Subgenomic	-0.67	-0.67	-0.7	-0.71	-0.65	-0.67	-0.68	-0.73
AUC OP	(-0.79, -0.51)	(-0.79, -0.51)	(-0.81, -0.55)	(-0.82, -0.57)	(-0.77, -0.48)	(-0.79, -0.5)	(-0.79, -0.52)	(-0.83, -0.59)
Swab N1	-0.65	-0.61	-0.65	-0.68	-0.62	-0.62	-0.65	-0.67
AUC OP	(-0.77, -0.47)	(-0.74, -0.42)	(-0.78, -0.48)	(-0.79, -0.52)	(-0.75, -0.43)	(-0.75, -0.44)	(-0.77, -0.48)	(-0.79, -0.51)
Swab	-0.58	-0.61	-0.58	-0.57	-0.63	-0.55	-0.55	-0.56
Subgenomic	(-0.73, -0.39)	(-0.75, -0.43)	(-0.72, -0.39)	(-0.72, -0.37)	(-0.76, -0.45)	(-0.71, -0.35)	(-0.71, -0.35)	(-0.71, -0.37)
log VL Day2	-0.58	-0.59	-0.59	-0.58	-0.59	-0.54	-0.53	-0.54
PostChal- lenge BAL	(-0.73, -0.39)	(-0.73, -0.4)	(-0.73, -0.4)	(-0.72, -0.38)	(-0.73, -0.39)	(-0.7, -0.34)	(-0.69, -0.32)	(-0.7, -0.34)
N1	-0.46	-0.32	-0.4	-0.43	-0.33	-0.36	-0.36	-0.39
log VL Day2	(-0.64, -0.24)	(-0.53, -0.08)	(-0.59, -0.17)	(-0.61, -0.21)	(-0.54, -0.09)	(-0.56, -0.13)	(-0.56, -0.13)	(-0.58, -0.16)
PostChal- lenge	-0.53	-0.39	-0.47	-0.5	-0.4	-0.44	-0.44	-0.47
OPswab N1	(-0.69, -0.33)	(-0.58, -0.15)	(-0.64, -0.25)	(-0.67, -0.29)	(-0.59, -0.17)	(-0.62, -0.21)	(-0.62, -0.21)	(-0.64, -0.25)
log VL Day2	-0.53	-0.39	-0.47	-0.5	-0.4	-0.44	-0.44	-0.47
PostChal- lenge	(-0.69, -0.33)	(-0.58, -0.15)	(-0.64, -0.25)	(-0.67, -0.29)	(-0.59, -0.17)	(-0.62, -0.21)	(-0.62, -0.21)	(-0.64, -0.25)
OPswab	-0.63	-0.6	-0.61	-0.62	-0.57	-0.56	-0.62	-0.62
Subgenomic	(-0.76, -0.46)	(-0.74, -0.41)	(-0.75, -0.43)	(-0.75, -0.44)	(-0.72, -0.38)	(-0.71, -0.37)	(-0.75, -0.44)	(-0.75, -0.44)
log VL Day2	-0.63	-0.6	-0.61	-0.62	-0.57	-0.56	-0.62	-0.62
PostChal- lenge	(-0.76, -0.46)	(-0.74, -0.41)	(-0.75, -0.43)	(-0.75, -0.44)	(-0.72, -0.38)	(-0.71, -0.37)	(-0.75, -0.44)	(-0.75, -0.44)
NasalSwab								
N1								

log VL Day2	-0.66	-0.62	-0.63	-0.63	-0.58	-0.57	-0.63	-0.64
PostChal-	(-0.78,	(-0.75,	(-0.76,	(-0.76,	(-0.73,	(-0.72,	(-0.76,	(-0.76,
lenge	-0.49)	-0.44)	-0.45)	-0.45)	-0.39)	-0.38)	-0.46)	-0.46)
NasalSwab								
Subgenomic								
